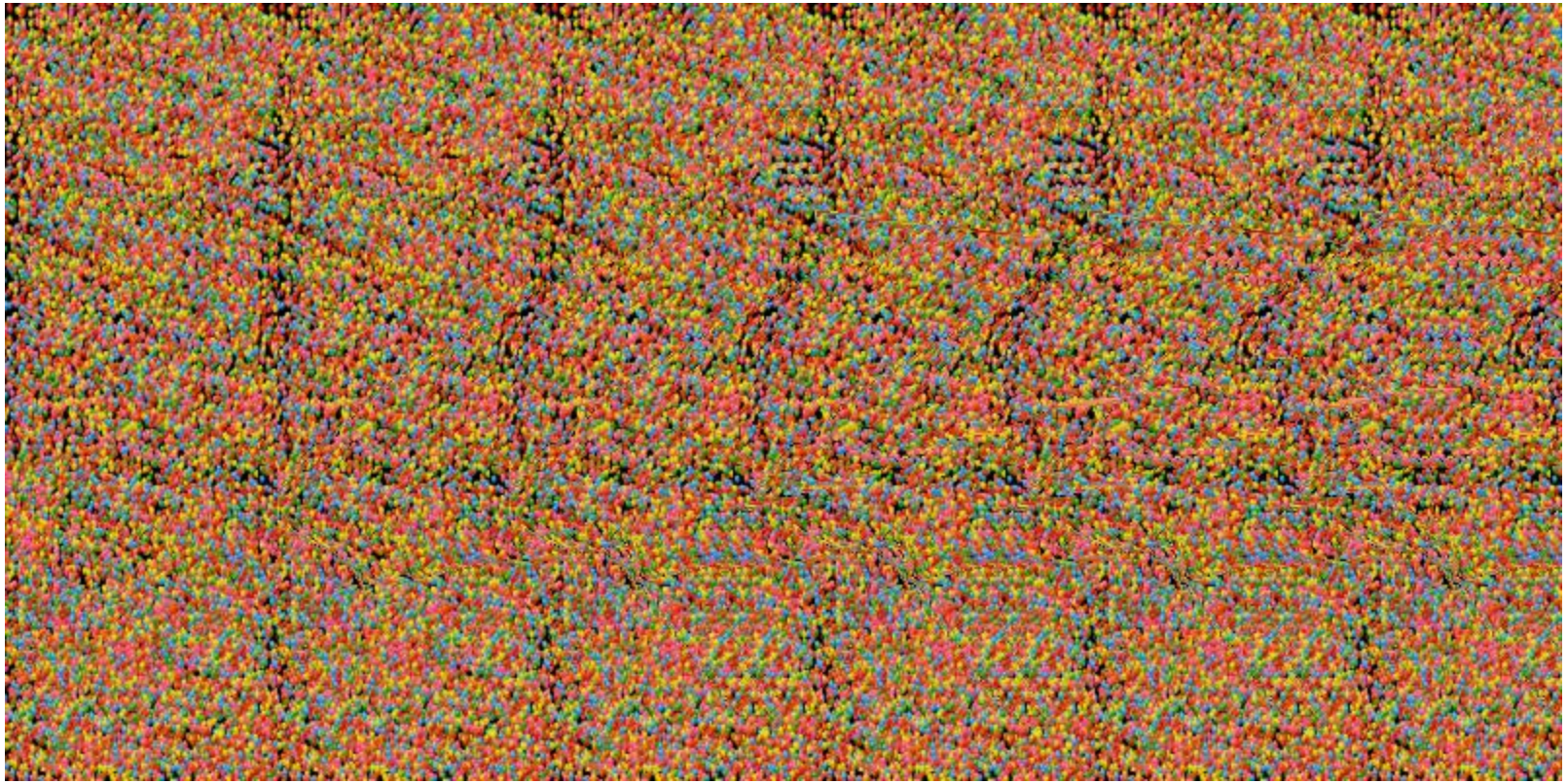
# CS5187: Vision and Image

## Binocular Stereo

What is this?



Single image stereogram,
https://en.wikipedia.org/wiki/Autostereogram



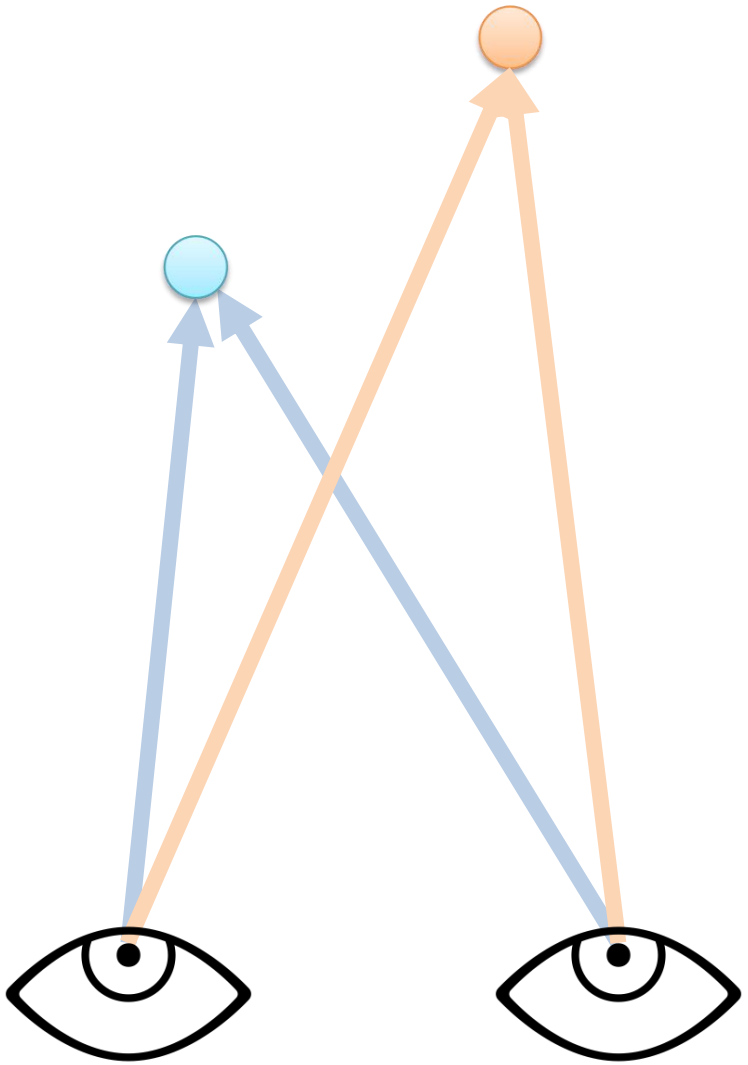
**“Mark Twain at Pool Table", no date, UCR Museum of Photography**



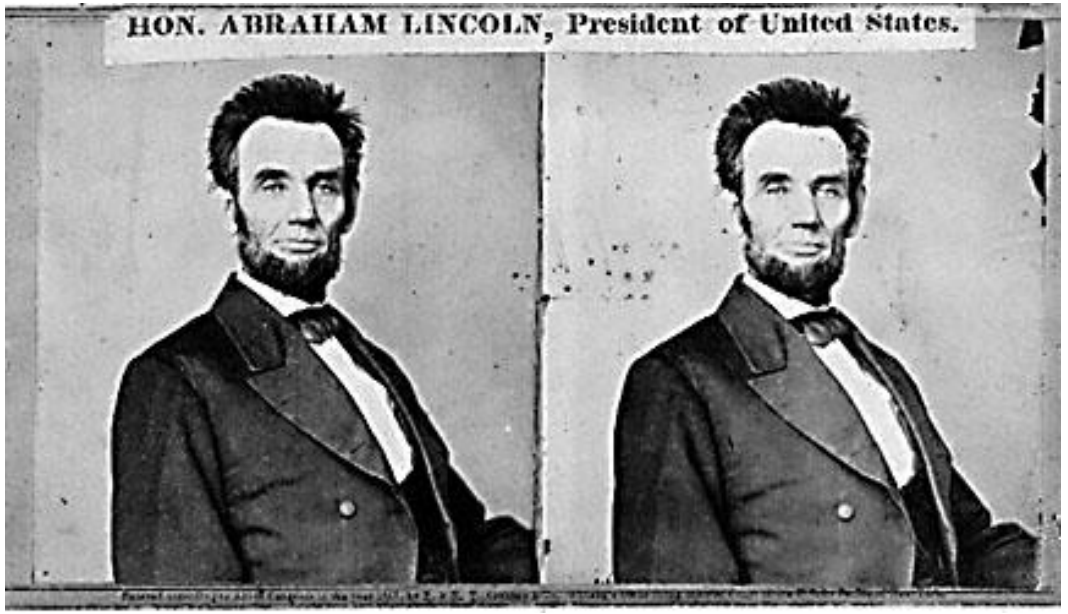
https://giphy.com/gifs/wigglegram-706pNfSKyaDug

# Stereo Vision as Localizing Points in 3D

- An object point will project to some point in our image
- That image point corresponds to a ray in the world
- Two rays intersect at a single point, so if we want to localize points in 3D we need 2 eyes



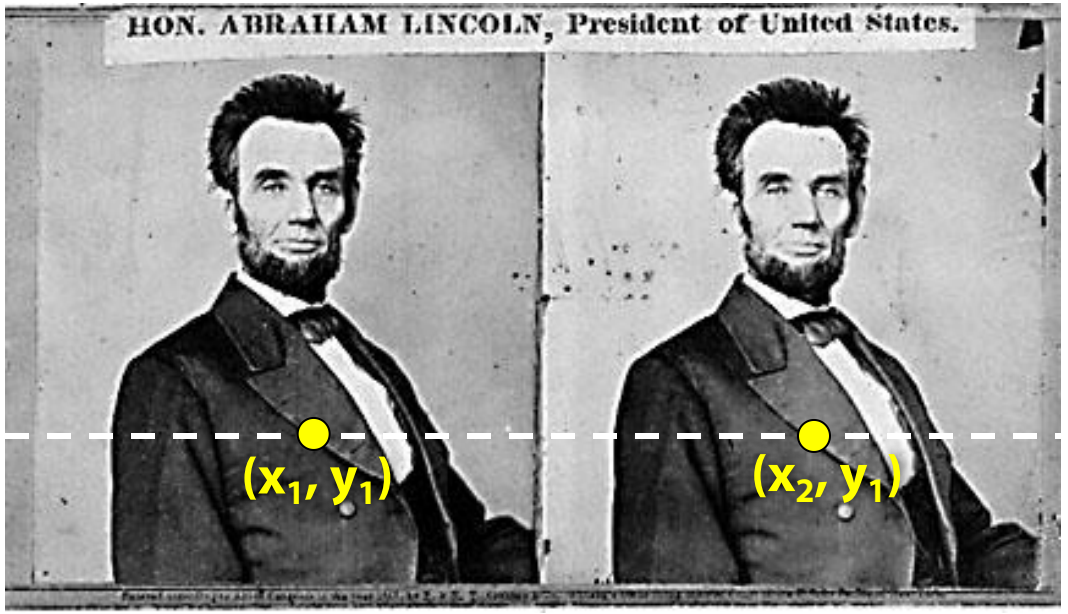
# Stereo




- Given two images from different viewpoints
  - How can we compute the depth of each point in the image?
  - Based on *how much each pixel moves* between the two images

# Epipolar geometry

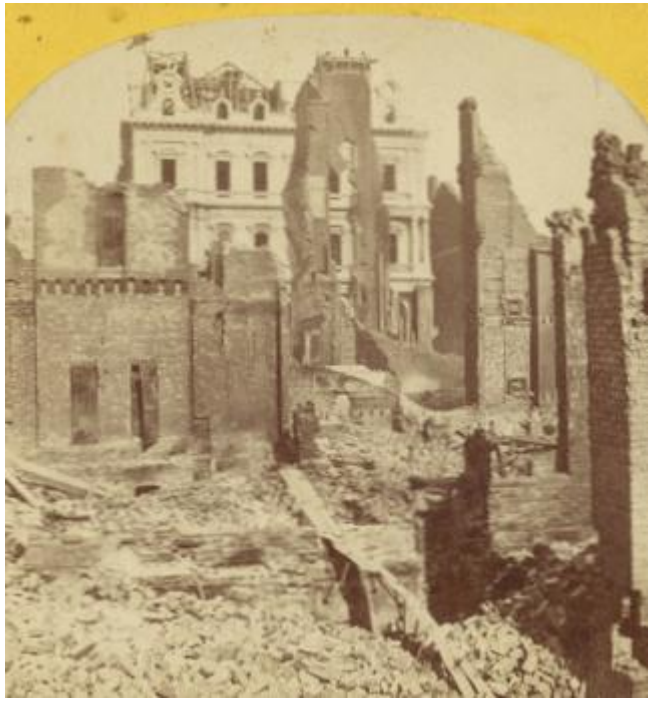
*epipolar lines*




Two images captured by a purely horizontal translating camera
(*rectified* stereo pair)

**$x_2 - x_1$ = the *disparity* of pixel $(x_1, y_1)$**

# Disparity = inverse depth



http://stereo.nypl.org/view/41729

(Or, hold a finger in front of your face and wink each eye in succession.)

# Your basic stereo matching algorithm

- **Match Pixels in Conjugate Epipolar Lines**
  - Assume brightness constancy
  - This is a challenging problem
  - Hundreds of approaches
    - A good survey and evaluation: http://www.middlebury.edu/stereo/